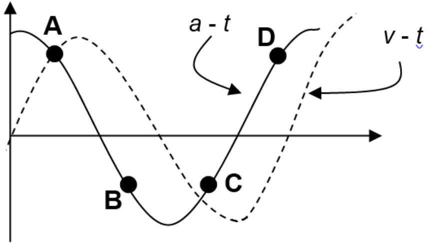
		$= ½ (100 \text{ mA} + 20 \text{ mA}) (8) =$ **480 mC**
19	**D**	At point P, the displacement, *x* is negative and velocity, *v* is negative. Since $a = -\omega^2 x \Rightarrow F = ma = -m\omega^2 x$, *x* negative will mean *F* is positive (B and C no longer plausible). Since *a-t* graph is a cosine graph, *v-t* graph will be a sine graph and hence A is not possible since its velocity is positive. 
20	**D**	Unpolarised light | After 1st polariser | After 2nd polariser